

SCENARIO BASED ASSESSEMENT

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COURSE NAME: OBJECT ORIENTED

PROGRAMMING IN C++

COURSE CODE: DSA0103

1. A hospital wants to estimate consultation charges for different patient categories. Design a C++ program for a Consultation Charge System. Design a C++ program for hospital management Task: A. Use function overloading to calculate charge for general consultation and specialist consultation. b. Use call by reference to modify the final bill amount. c. Use default arguments to add a standard service charge if needed. d. Use function prototyping for all calculation functions. e. Display the patient type and totalConsultation charge.(BL3)

```
main.cpp
8
9 #include <iostream>
10 using namespace std;
11
12 // Function Prototypes
13 double calculateCharge(double fee);
14 double calculateCharge(double fee, double specialistCharge);
15 void applyDiscount(double &bill, double discount);
16 double addServiceCharge(double bill, double serviceCharge = 100);
17
18 int main()
19 {
20     int choice;
21     double fee, specialistCharge, discount, totalBill;
22
23     cout << "==== Consultation Charge System =====\n";
24     cout << "1. General Consultation\n";
25     cout << "2. Specialist Consultation\n";
26     cout << "Enter Patient Type (1/2): ";
27     cin >> choice;
28
29     if(choice == 1)
30     {
31         cout << "Enter General Consultation Fee: ";
32         cin >> fee;
33
34         totalBill = calculateCharge(fee);
```

input

```
Enter Patient Type (1/2): 1
Enter General Consultation Fee: 500
Enter Discount Amount: 50

----- Bill Details -----
Patient Type      : General
Total Charge     : 550

...Program finished with exit code 0
Press ENTER to exit console.
```

OUTPUT:

```
==== Consultation Charge System =====
1. General Consultation
2. Specialist Consultation
Enter Patient Type (1/2): 1
Enter General Consultation Fee: 500
Enter Discount Amount: 50

----- Bill Details -----
Patient Type      : General
Total Charge     : 550

...Program finished with exit code 0
Press ENTER to exit console.
```

2. Design a C++ program Fitness Centre Membership System. A gym wants to manage member subscriptions, track plans, and calculate membership fees.

Task:

Create a class Member with attributes: Member ID, Name, Plan Type, and Fee. Implement member functions to assign plans, calculate fees, and display membership details. Use an array of objects to manage multiple members. Add validation for valid membership plans. Include a static member to track total active members. Evaluate how your design supports easy modification of pricing plans.

SOLUTION:

```
main.cpp
8
9 #include <iostream>
10 #include <string>
11 using namespace std;
12
13 class Member
14 {
15 private:
16     int memberID;
17     string name;
18     string planType;
19     double fee;
20
21 public:
22     static int totalMembers;
23
24     void assignPlan();
25     void calculateFee();
26     void display();
27 };
28
29 int Member::totalMembers = 0;
30
31 // Assign Member Details
32 void Member::assignPlan()
33 {
34     cout << "\nEnter Member ID: ";
35     cin >> memberID;
```

input

```
----- Member Details -----
Member ID   : 0
Name        :
Plan Type   : Basic
Fee         : Rs. 1000
Total Active Members : 3
```

OUTPUT:

```
input
----- Fitness Centre Membership System -----
Enter Number of Members: 3

Member 1
Enter Member ID: 101
Enter Member Name: SHRUTHI
Enter Plan Type (Basic/Silver/Gold): GOLD
Invalid Plan! Basic Plan Assigned.

Member 2
Enter Member ID: 102
Enter Member Name: HARINI
Enter Plan Type (Basic/Silver/Gold): SILVER
Invalid Plan! Basic Plan Assigned.

Member 3
Enter Member ID: KAVI
Enter Member Name: Enter Plan Type (Basic/Silver/Gold): Invalid Plan! Basic Plan Assigned.

----- Membership Details -----
----- Member Details -----
Member ID   : 101
Name        : SHRUTHI
Plan Type   : Basic
Fee         : Rs. 1000
```

3. A bank wants to calculate loan repayment details for customers. Design a C++ program for a Loan Repayment System. Task: A. Use function prototyping to declare EMI-related functions. b. Use default arguments for default interest rate where applicable. c. Apply call by reference to update payable amount after subsidy. d. Use inline functions for simple interest calculations. e. Display the loan amount, interest, and repayment details clearly.

SOLUTION:

```
main.cpp
1
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8
9 #include <iostream>
10 using namespace std;
11 inline double calculateInterest(double loanAmount, double rate, int years);
12 double calculatePayable(double loanAmount, double interest);
13 void applySubsidy(double &payableAmount, double subsidy);
14 void displayDetails(double loanAmount, double interest, double payableAmount);
15
16 int main()
17 {
18     double loanAmount, rate = 10.0, interest, payableAmount, subsidy;
19     int years;
20
21     cout << "==== Loan Repayment System =====< endl;
22
23     cout << "Enter Loan Amount: ";
24     cin >> loanAmount;
25
26     cout << "Enter Loan Period (Years): ";
27     cin >> years;
28
29     cout << "Enter Interest Rate (Press 0 for Default 10%): ";
30     cin >> rate;
31
32     if(rate == 0)
33         rate = 10.0; // Default Argument Value
34
35     calculateInterest(loanAmount, rate, years, interest);
36     payableAmount = calculatePayable(loanAmount, interest);
37     applySubsidy(payableAmount, 5000);
38     displayDetails(loanAmount, interest, payableAmount);
39
40     return 0;
41 }
```

input

```
----- Loan Details -----
Loan Amount      : 10000
Interest Amount  : 2000
Final Payable    : 7000
```

OUTPUT:

```
===== Loan Repayment System =====
Enter Loan Amount: 10000
Enter Loan Period (Years): 2
Enter Interest Rate (Press 0 for Default 10%): 0
Enter Subsidy Amount: 5000

----- Loan Details -----
Loan Amount      : 10000
Interest Amount  : 2000
Final Payable    : 7000

...Program finished with exit code 0
Press ENTER to exit console.
```

4. A delivery company wants to automate the calculation of parcel delivery charges based on parcel weight, delivery distance, and delivery type. Design and implement a C++ program using functions and object-oriented programming concepts to develop a Parcel Charge Management System. a. Create a class Parcel with appropriate private data members such as parcel type, weight, distance, and packaging charge. b. Use public member functions to read parcel details, calculate the delivery charge, and display the parcel information. c. Use function overloading to calculate charges for Normal Delivery and Express Delivery. d. Use default arguments to assign a default packaging charge when it is not specified by the user. e. Use an inline function to calculate the base charge based on parcel weight and delivery distance. f. Maintain the total number of parcels processed using a static data member and display it through a static member function.

SOLUTION:

```
main.cpp
8
9 #include <iostream>
10 #include <string>
11 using namespace std;
12
13 class Parcel
14 {
15 private:
16     string parcelType;
17     float weight;
18     float distance;
19     float packagingCharge;
20     float totalCharge;
21
22     // Inline Function
23     inline float baseCharge()
24     {
25         return weight * distance * 2;
26     }
27
28 public:
29     static int totalParcels;
30
31     void readDetails();
32     void calculateCharge();
33     void calculateCharge(float expressCharge); // Function Overloading
34     void display();
35
36     ~Parcel() { totalParcels--; }
37 };
38
39 int Parcel::totalParcels = 0;
40
41 int main()
42 {
43     Parcel p;
44     p.readDetails();
45     p.calculateCharge();
46     p.display();
47     return 0;
48 }
```

input

```
Enter Weight (kg): 8
Enter Distance (km): 150
Enter Packaging Charge (Press 0 for Default 50): 0

Delivery Type
1. Normal Delivery
2. Express Delivery
Enter Choice: 1
```

```
===== Parcel Charge Management System =====
Enter Number of Parcels: 2

Parcel 1

Enter Parcel Type: DOCUMENTS
Enter Weight (kg): 5
Enter Distance (km): 100
Enter Packaging Charge (Press 0 for Default 50): 0

Delivery Type
1. Normal Delivery
2. Express Delivery
Enter Choice: 1

Parcel 2

Enter Parcel Type: ELECTRONICS
Enter Weight (kg): 8
Enter Distance (km): 150
Enter Packaging Charge (Press 0 for Default 50): 0

Delivery Type
1. Normal Delivery
2. Express Delivery
Enter Choice: 
```

5. A utility company wants to automate the calculation of water bills for domestic and commercial consumers. Design and implement a C++ program using functions and object-oriented programming concepts to develop a Water Billing System. create a class Consumer with appropriate private data members such as consumer ID, consumer type, water consumption (in liters), and bill amount. b. Use public member functions to input consumer details, calculate the water bill, and display the billing information. c. Implement function overloading to calculate water bills for Domestic and Commercial consumers using different tariff rates. d. Use call by reference to update the bill amount after applying a government subsidy. e. Use default arguments to include a service tax, where a default tax rate is applied unless specified. f. Declare all functions using function prototyping before their definitions. g. Maintain the total number of consumers processed using a static data member and display it through a static member function. h. Create an array of objects to store and process the details of multiple consumers, and display the consumer type, water consumption, subsidy-adjusted bill amount.

SOLUTION:

```
main.cpp
8
9 #include <iostream>
10 using namespace std;
11
12 // Function Prototypes
13 double calculateBill(double liters);           // Domestic
14 double calculateBill(double liters, double rate); // Commercial
15 void applySubsidy(double &bill, double subsidy); // Call by Reference
16 double addServiceTax(double bill, double tax = 50); // Default Argument
17
18 class Consumer
19 {
20 private:
21     int consumerID;
22     string consumerType;
23     double waterConsumption;
24     double billAmount;
25
26 public:
27     static int totalConsumers;
28
29     void input();
30     void calculate();
31     void display();
32     static void displayTotalConsumers();
33 };
34
35 // Initializing Static Data Member
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